

Individual Contribution Journal

For this experiment, I independently completed and checked all the cells in the `Module_03_Lab_Exercise.ipynb` notebook and completed the full data analysis process. I personally loaded the Wine dataset and performed exploratory data analysis (EDA), including checking the data shape, class distribution, and correlation matrix of the first six features. I used Pandas, Matplotlib, and Seaborn to complete these tasks and make sure the data was clean and reliable.

During the model building process, I completed the six main steps of machine learning, including data preparation, feature selection, stratified data splitting with an 80/20 ratio, and training logistic regression and decision tree models. During the hands-on session, I also tested different combinations of chemical features by myself. I compared the results and studied how different features affected classification accuracy. I also examined the confusion matrices and classification reports to better understand the results of each model.

For the theoretical assessment, I independently completed the questions about the three main types of machine learning: supervised learning, unsupervised learning, and reinforcement learning. I answered all of the questions correctly. Finally, I organized all the experimental results, adjusted the page layout, and completed the final PDF export and formatting check based on the course requirements.